

Lily Gebhart

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Education

Stanford University

Neurosciences PhD Student

Advised by Andrea Goldstein-Piekarski

Palo Alto, CA

Sept. 2025 - Present

Occidental College

B.A. Cognitive Science, B.A. Mathematics

Graduated summa cum laude with minors in Neuroscience, Computer Science

Los Angeles, CA

Aug. 2021 - May 2025

Awards & Fellowships

April 2026	Department of Energy Computational Science Graduate Fellowship (DOE CSGF)
April 2026	National Science Foundation Graduate Research Fellowship Program (NSF GRFP) [Declined]
May 2025	Mathematics Department Excellence Award
May 2025	Cognitive Science Outstanding TA/ Peer Mentor Award
Sept. 2024	Phi Beta Kappa Honor Society Member: One of ten early inductees from the 2025 graduating class.
Mar. 2024	Barry Goldwater Scholarship
Apr. 2022	General Chemistry Award
2021-2025	Margaret Bundy Scott Scholarship: A \$120,000 merit scholarship for study at Occidental College

Publications

- [1] Gebhart, L. C., Levitan, C. A., & Motala, A. (2025). A User-Friendly Framework for Short Duration Visual Time Perception. *PsychArXiv*.
- [2] Gebhart, L. & Li, J. (2025). Integrated Co-Occurrent and Semantic Mechanisms for Long Term Memory Retrieval. *International Conference for Cognitive Modeling 2025*.
- [3] Omelyusik, V., Gebhart, L., & Nair, S. (2025). Communication in Brain Circuits and Systems: A Primer. *IEEE International Symposium on Circuits and Systems 2025*.
- [4] Gebhart, L. & Kolassa, J. (2024). Adjustments for Kurtosis and Continuity on the Prentice Test. *Advances in Pure Mathematics*, 14(2), 101-117.
- [5] Gebhart, L. & Snowden-Ifft, D. (2023). Muon Tomography for Detection of Dynamic Border Tunnels, *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment*, 1053, 168383.

Published Abstracts

- [1] Marks, A., Bobrovnikov, E., Gebhart, L. C., Clarke, J. L., Demayo, C., Taylor, J. W., Paulsen, K., Stallard, S. & Chang, S. M. (2025). An intervention in disguise: An immersive care model for families where a parent has a brain tumor. *American Society of Clinical Oncology Proceedings*.
- [2] Gebhart, L. C., Marks, A. L., Demayo, C., Clarke, J. L., Paulsen, K., Taylor, J. W., & Chang, S. (2025). Milton Marks Neuro-Oncology Family Camp (MMFC): 10 Years of Community Building and Support for Families with a Parent Diagnosed with Brain Cancer. *Society for Neuro-Oncology Annual Meeting Proceedings*.

Oral Presentations

- [1] Gebhart, L. (July 2025) Integrated Co-occurent and Semantic Mechanisms for Modeling Long Term Memory Retrieval. International Conference on Cognitive Modeling. Ohio State University, Columbus, OH, United States.
- [2] Gebhart, L. (Apr. 2024) The Effect of Love on Durational Time Perception. Occidental College Cognitive Science Department Advanced Research Talks. Occidental College, Los Angeles, CA, United States.
- [3] Gebhart, L. (Jan. 2024) Adjustments for Kurtosis and Continuity on the Prentice Test. National Conference for Undergraduate Women in Mathematics. University of Nebraska - Lincoln, Lincoln, NE, United States.
- [4] Gebhart, L. (Nov. 2023) Adjustments for Kurtosis and Continuity on the Prentice Test. 2023 Southern California Conference for Undergraduate Research. California State University Fullerton, Fullerton, CA, United States.
- [5] Gebhart, L. (Nov. 2023) Adjustments for Kurtosis and Continuity on the Prentice Test. Occidental College Math Department Student REU Talks. Occidental College, Los Angeles, CA, United States.

- [6] **Gebhart, L.** (Jul. 2023) Adjustments for Kurtosis and Continuity on the Prentice Test. 2023 Discrete Mathematics and Theoretical Computer Science (DIMACS) NSF REU Program Conference. Rutgers University, Piscataway, NJ, United States.
- [7] **Gebhart, L.** (Nov. 2022) Simulated Muography for Detection of Dynamic Border Tunnels. 2022 Southern California Conference for Undergraduate Research. Pepperdine University, Malibu, CA, United States.
- [8] **Gebhart, L.** (Aug. 2022) Simulated Muography for Detection of Dynamic Border Tunnels. 2022 Summer Research Program Conference. Occidental College, Los Angeles, CA, United States.

Poster Presentations

- [1] **Gebhart, L.** (May 2025) Communication in Brain Circuits and Systems: A Primer. IEEE International Symposium on Circuits and Systems. London, England.
- [2] **Gebhart, L.** (Nov. 2024) Integrated Co-occurrent and Semantic Mechanisms for Modeling Long Term Memory Retrieval. 2024 Southern California Conference for Undergraduate Research. California State University San Bernadino, San Bernadino, CA, United States.
- [3] **Gebhart, L.** (Aug. 2024) Integrated Co-occurrent and Semantic Mechanisms for Modeling Long Term Memory Retrieval. 2024 Summer Research Program Conference. Occidental College, Los Angeles, CA, United States.
- [4] **Gebhart, L.** (Apr. 2024) Adjustments for Kurtosis and Continuity on the Prentice Test. National Conference for Undergraduate Research. California State University Long Beach, Long Beach, CA, United States.

Research Funding & Training Programs

Missouri Neuroscience Research Experience for Undergraduates (REU)

University of Missouri, MO

Supported by NSF Grant CNS-1950787

May 2024 - Present

Funding by the National Science Foundation for a 7 month part-time remote research fellowship at the University of Missouri-Columbia, MO.

Science Scholars Fellowship

Occidental College, CA

Supported by the Fletcher-Jones Foundation

Jan. 2024 - Present

A selective \$15,000 scholarship grant supporting 15 months of research during the Spring 2024, Fall 2024, and Spring 2025 semesters and participation in the Occidental College Summer Research Program.

Discrete Mathematics and Theoretical Computer Science NSF REU

Rutgers University, NJ

Supported by NSF Grant CNS-2150186

May 2023 - Jul. 2023

Funding by the National Science Foundation for a 3 month full-time summer research fellowship at Rutgers University, NJ.

Academic Student Project Travel Grants

Occidental College, CA

Supported by the Occidental College Undergraduate Research Center

Dec. 2023, Mar. 2024

Funding for travel and registration expenses related to my attendance of the 2024 National Conference for Undergraduate Women in Mathematics at the University of Nebraska-Lincoln, NE and the 2024 National Conference for Undergraduate Research in Long Beach, CA.

Student Research Assistantship Program Fellowships

Occidental College, CA

Supported by the Undergraduate Research Center Funds at Occidental College

Spring, Fall 2023

A \$750 research grant supporting 50 hours of research during the semester for the projects, "The Effect of Love on Durational Time Perception" (Fall 2023) and "Integrated Mechanisms for Long Term Memory Retrieval" (Spring 2023).

Academic Student Project Grants

Occidental College, CA

Supported by the Occidental College Undergraduate Research Center

Aug. 2023, Jan. 2024

Two \$300 grants for funding participant compensation for the project, "The Effect of Love on Durational Time Perception".

Research Early Access Program

Occidental College, CA

Supported by the Sherman Fairchild Foundation

May 2022 - Aug. 2022

Funding for a 3 month full-time summer research fellowship at Occidental College, CA.

Teaching & Professional Experience

Research Assistant

Remote

Synaptic Insights Consulting

Sept. 2025 - Present

Assist in data visualization, analytics, communication, and dissemination for human flourishing and social connection research projects.

Program Co-Coordinator and Mathematics Tutor

Scientific Scholars Achievement Program

Occidental College, CA

Nov. 2022 - May 2025

- Oversaw, interviewed, hired, and trained a body of 20 tutors serving students in introductory courses in seven STEM disciplines.
- Introduced initiatives to increase attendance and boost SSAP's campus presence by hosting beginning of semester welcome events, introducing attendance tracking measures, redesigning training, and developing targeted advertising initiatives.
- Assisted students with homework and exam preparation of introductory and advanced undergraduate mathematics courses twice weekly.

Teaching Assistant

Occidental College

Occidental College, CA

Jan. 2023 - May 2025

- Computational Neuroscience (Spring 2024, Spring 2025)
- Fundamentals of Computer Science (Spring 2023)
- Data Structures (Spring 2023)
- Ordinary Differential Equations Grader (Fall 2021)

Curriculum Assistant

Boundless Brilliance

Remote

Jun. 2023 - Aug. 2023

- Restructured and updated the organization's body of 300+ experiments and activities for presentations to students in grades K-6.
- Illustrated a STEM coloring book, and developed 12 summer STEM packets and workshop series for college chapters.

Committees & Service

Stanford Neurosciences IDP Student Program Committee

Committee Member

Stanford, CA

Jan. 2026 - Present

Stanford Neuroscience Application Assistance Program (SNAAP)

Leadership Team Member

Stanford, CA

Sep. 2025 - Present

Camp Kaleidoscope

Youth Counselor

Stanford, CA

Nov. 2025 - Present

Milton Marks Family Neuro-Oncology Camp

Board Member, Youth Counselor

San Francisco, CA

Apr. 2025 - Present

Boundless Brilliance: 501(c)3 Nonprofit

Nonprofit Student Board Member, Board Member

Los Angeles, CA

Nov. 2022 - Present

Boundless Brilliance: Occidental College Chapter

Chapter President, Chapter Outreach Co-Coordinator

Occidental College, CA

May 2022 - May 2025

Skills

Programming

Python, R, MATLAB, Java

Drawing & Typesetting

Microsoft Suite, L^AT_EX, Adobe Creative Suite, Canva